



Firearm Violence and Legislation in the United States

Data Visualization Report

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Data Visualization — U-tad

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1 Introduction

This project explores the relationship between state firearm legislation and gun violence in the United States using data spanning 1976–2023. While estimated household gun ownership is included as a contextual variable, it is not treated as the primary explanatory factor. Evidence from countries such as Finland and Switzerland where high gun ownership coexists with low rates of gun violence, suggests that the regulatory environment surrounding firearms may matter more than ownership prevalence alone. This project therefore focuses on whether the strength and type of state firearm laws is associated with lower rates of firearm homicide, and whether changes in legislation over time correspond to measurable shifts in a state's gun violence standing.

1.1 Methodology

The project followed the process below to achieve the goal described above:

1. Dataset selection and join strategy design
2. Data analysis outside Tableau
3. Research questions derived from data exploration
4. Visualization types needed to answer each question and storytell
5. Data source connection, relationships, and calculated fields in Tableau
6. Tableau feature selection (parameters, sets, filters, actions, tooltips)
7. Dashboard design and interactive storytelling
8. Use of AI assistance for laborious or repetitive tasks (see Section 1.3)

1.2 Sources

Numerous sources are used throughout the project such as:

- Official dataset documentation and codebooks^{1,2}
- Harvard Dataverse and Tufts CTSI dataset metadata^{3,2}
- Academic literature on firearm ownership proxies^{4,3}
- Academic literature on gun law effects^{5,6,7}
- Tableau official documentation⁸
- Lecture notebooks and course materials
- AI assistance (declared in Section 1.3)

Regardless of the source, these are indicated through bibliography and references, or in the case of AI, an explicit note indicating the model and prompt used.

1.3 AI Usage Declaration

AI assistance was used for laborious, repetitive, or mechanical tasks throughout this project. All AI output was reviewed critically before use. Analytical conclusions, Tableau visualizations, and interpretation of results were not produced by AI. The model used in all cases was **Claude Sonnet 4.6**, accessed via the Perplexity AI assistant. Detailed prompts and responses are documented inline at each point of use; the table below provides a summary index.

Task	Model	Section
Dataset research and pairing strategy	Claude Sonnet 4.6	2
State-abbreviation calculated field	Claude Sonnet 4.6	4.1
Document labelling and cross-referencing	Claude Sonnet 4.6	1.3.1
Report structuring and LaTeX formatting	Claude Sonnet 4.6	1.3.2
Supplementary chart generation	Claude Sonnet 4.6	4.3

Table 1: AI usage index

1.3.1 Document Labelling and Cross-Referencing

Adding `\label` tags to named sections and propagating `\ref` calls consistently throughout a long document is mechanical and error-prone when done manually. Modern LLMs are well-suited to this kind of structural text analysis, as they can parse the full document, identify every mention of a concept, and apply uniform markup in a single pass.

Prompt: *Add labels to questions Q1, Q2 and Q3 in the document and reference those labels whenever those questions are referenced in the document.*

Response summary: Added `\label{q1}`, `\label{q2}`, `\label{q3}` to the three `\subsubsection` headings, inserted `\sref{q1/q2/q3}` at the end of each item in the numbered question list, and prefixed the corresponding Conclusions bullets and Visualization Summary table rows with the matching `\sref` call, creating clickable hyperlinks throughout the PDF.

1.3.2 Report Structuring and LaTeX Formatting

AI was used to assist with LaTeX boilerplate, section organisation, consistent page-break placement, and bibliography management — tasks that are time-consuming but do not require analytical judgement.

Prompt: *For visualizations, make it so each of them is on a different page alongside with their text section, so page breaks are consistent and it doesn't break in weird places.*

Response summary: Added `\usepackage{float}` to the preamble, inserted `\newpage` before each visualization subsection (Q1, Q2, Q3, Dashboards, Summary), and changed all figure placement specifiers from `[h]` to `[H]` to force figures to appear exactly at their source position rather than floating.

2 Datasets

Two public datasets were selected and used in conjunction for this project. Both are accessible through their corresponding links and are available in different formats. The first dataset was identified independently; finding a compatible second dataset was AI-assisted (Claude Sonnet 4.6; see Section 1.3).

Prompt: *I already have a state-year level dataset on estimated household gun ownership in the US. I need a second dataset that contains information about firearm legislation on a state-year basis so I can join them easily. What publicly available datasets would you recommend?*

Response summary: Recommended the Tufts CTSI State Firearm Law Database, noting that it tracks 72 binary law provisions per U.S. state per year from 1976 to 2024, shares the same state-year grain as the ownership proxy dataset, and is widely used in public health research, making it a straightforward join candidate.

2.1 Dataset 1: Firearm Suicide Proxy for Household Gun Ownership, 1949–2023

The first dataset¹ is a state-year level proxy for household gun ownership in the United States. Direct gun ownership data is not publicly available at the state level, so this dataset uses the firearm suicide proxy methodology: the proportion of suicides committed with a firearm (FSS) is used as an indirect estimate of household gun prevalence by state and year. It is hosted on Harvard Dataverse.

- **Source:** Harvard Dataverse, Kang et al.³
- **Grain:** One row per U.S. state per year
- **Key columns:** State, Year, FSS (firearm suicide share), firearm suicides, total suicides, firearm homicide rate, total homicides, population
- **Date span:** 1949–2023 (75 years, 50 states + D.C.)
- **Format:** Comma-separated values (.csv)

This dataset was chosen because it provides a long term, population adjusted, and academically validated measure of gun prevalence that can be compared across states and over time. FSS is not a literal count of guns per household but an indirect proxy widely used in public health and policy research.⁴ FSS is retained in this project as a contextual indicator of estimated gun prevalence on the scatter plot; firearm homicide rate serves as the primary outcome measure throughout.

2.2 Dataset 2: State Firearm Law Database, 1976–2024

The second dataset² is maintained by the State Firearm Law Research Group at Tufts University and tracks the presence or absence of 72 firearm safety law provisions across all 50 U.S. states for every year from 1976 to 2024.

- **Source:** Tufts CTSI — State Firearm Law Database²
- **Grain:** One row per U.S. state per year
- **Key columns:** State, Year, 72 binary law indicator columns (e.g. permit required, background check, waiting period, stand-your-ground, child access prevention)
- **Date span:** 1976–2024
- **Format:** Excel (.xlsx)

This dataset was chosen because it provides a historically accurate, year-by-year record of state firearm legislation, enabling analysis of whether law changes correspond to changes in gun violence rates. It is non partisan by design and is widely used in academic public health research.²

2.3 Join Strategy

Both datasets were connected in Tableau using a **relationship** on the shared keys `state` and `year`, configured with **many-to-many** cardinality and **some records match** referential integrity on both sides. Many-to-many cardinality is appropriate here because both tables share the same grain (one row per state per year) and neither side acts as a unique dimension table. The “some records match” setting instructs Tableau not to assume all keys are present in both tables, handling non-overlapping rows safely without discarding data prematurely. Tableau resolves the relationship by performing aggregations on each table independently before combining results, preventing row duplication. The overlapping date range of both datasets yields an effective analysis window of **1976–2023**.

2.4 Analysis

Prior to connecting the data in Tableau, both files were inspected to understand structure, null patterns, and label consistency. Key observations:

- Both datasets use identical full state name format (e.g. Alabama, North Carolina), allowing for a join on state
- The ownership proxy file includes several pre calculated rate columns derived from raw counts and population
- The law database contains 72 binary columns but it also includes a total law count column (`lawtotal`) that can be used directly
- Both datasets include a sufficient overlapping year window between 1976–2023 that can be used to join

2.5 Processing

Both source files were connected to Tableau without modification to the raw data. No column renaming, deduplication, or external pre-processing was required. The only structural adjustment was the exclusion of the District of Columbia via a Tableau filter, as it is present in the ownership proxy dataset but absent from the state firearm laws dataset, which would have prevented a valid join for that entity. All visualizations are filtered to the years 1976–2023, the window for which both datasets have complete data.

2.6 Research Questions

Three questions were formulated based on data exploration and storytelling intent:

1. Is a state's firearm homicide rate associated with the number and type of firearm laws it has in place?^{2.6.1}
2. How have firearm homicide rates evolved over time depending on the firearm laws passed in each state, whether measured by the quantity of laws or by the categories of law types enacted?^{2.6.2}
3. For a specific state, does passing more firearm laws correspond to a reduction in firearm homicide rates, and how does this affect its ranking relative to other states?^{2.6.3}

2.6.1 Question 1: Homicide Rates and the Law Landscape

This is the entry point of the analysis. Before drawing any temporal or causal conclusions, it is necessary to establish whether a cross-sectional pattern exists: do states that have enacted more laws or certain types of laws tend to have lower firearm homicide rates? FSS is included as a contextual variable to show where estimated gun ownership sits in this picture, without treating it as an outcome or a direct cause.

2.6.2 Question 2: Homicide Trends by Law Profile Over Time

If a cross-sectional association exists, the natural follow-up is whether it has strengthened or weakened over time. By grouping states according to their law profile, either by quantity tier or by law type category (see Section 3.2), it becomes possible to observe whether homicide trajectories have diverged, and whether that divergence accelerated after key legislative periods such as the early 1990s.

2.6.3 Question 3: State-Level Law History and Homicide Standing

Q1 and Q2 analyze the broader picture; Q3 brings the analysis down to the individual state, making it personally relevant and explorable for any viewer. The motivation is to let someone examine their own state: when did it pass laws, how many, and does its current homicide ranking reflect that legislative history?

3 Visualization Prep

Before building charts in Tableau, design and structural decisions were made to align visualizations with the storytelling intent.

3.1 Storytelling

The project follows a four act structure. Act 1 establishes the geographic and cross sectional baseline (where is violence highest, what does the law landscape look like). Act 2 adds the temporal dimension, showing how homicide rates have diverged by law profile over 1976–2023, using both categorization schemes defined in Section 3.2: quantity tier and law type group (Popular vs. Evidence-Based). Act 3 narrows to individual states, tracing each state’s legislative history against its homicide ranking relative to all others. Act 4 zooms into the selected law provisions directly, comparing the average homicide trend for the Popular and Evidence Based groups to assess which shows the more consistent association with changes in homicide rates.

3.2 Law Categorization

Two complementary categorization schemes are used across the visualizations to group states by their law profile. Both are applied through a shared parameter that lets the viewer toggle between them.

By quantity — states are assigned to one of three tiers based on their total law count (`lawtotal`) for a given year:

- **Low** — states with a relatively small number of laws in effect
- **Medium** — states in the middle tier
- **High** — states with a large number of laws in effect

By type — rather than comparing all 72 laws, a focused subset was selected: the three most widely adopted laws (“Popular”) and the three laws with the strongest evidence base in the literature (“Evidence-Based”). States are then classified into four mutually exclusive categories based on whether they have enacted two or more laws from either group:

- **Weak** — fewer than two Popular laws and fewer than two Evidence-Based laws
- **Popular** — two or more Popular laws, but fewer than two Evidence-Based laws
- **Evidence** — two or more Evidence-Based laws, but fewer than two Popular laws
- **Both** — two or more laws from each group

3.3 Colors and Themes

Color choices were made deliberately to minimize viewer bias and maximize clarity:

- **Neutral colors** were used throughout to avoid leading the viewer toward a political interpretation. Grey tones were applied to states with no meaningful change in laws or with a weak law profile, so that they recede visually without being misleading.
- **Orange gradient** — a somewhat neutral but gravity-suggesting tone — was chosen for value-encoding gradients (e.g. firearm homicide rates on maps and scatter plots), where darker shades indicate higher rates.
- **Gridlines** were kept minimal to reduce visual noise, with color used as the primary channel for directing the viewer’s attention to data marks rather than chart furniture.

4 Visualizations

All visualizations share a consistent color scheme as described in Section 3.3 and use the law categorization schemes defined in Section 3.2.

4.1 Q1 – Homicide Rates and the Law Landscape

Two charts answer Q1^{2.6.1}. The first is a **choropleth map** (Figure 1) displaying the firearm homicide rate across all U.S. states for a user-selected year, encoded with an orange gradient so darker states indicate higher rates. A **parameter** drives the year selection, and a **calculated field** supplies the two-letter abbreviation labels shown on each state mark.

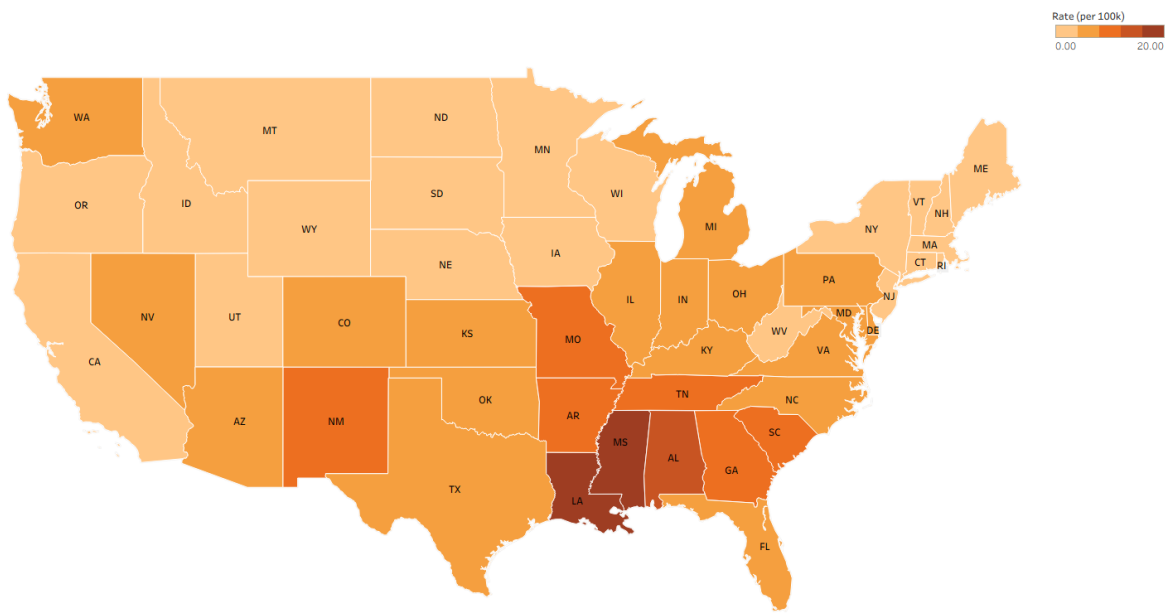
Generating the state abbreviation calculated field was AI-assisted (Claude Sonnet 4.6; see Section 1.3), as writing all 50 WHEN clauses by hand is purely mechanical and error-prone. The prompt and response are shown below (Prompt 4.1).

Prompt: Generate a Tableau calculated field using a CASE statement that maps each U.S. state's full name to its two-letter abbreviation for all 50 states. Only output the code.

Response:

```
CASE [State]
WHEN "Alabama" THEN "AL"
WHEN "Alaska" THEN "AK"
...
WHEN "Wisconsin" THEN "WI"
WHEN "Wyoming" THEN "WY"
END
```

Firearm Homicide Rate by State (2023)



Map based on Longitude (generated) and Latitude (generated). Colour shows average of Firearm Homicide Rate. The marks are labelled by CF StateAbbreviation. Details are shown for CF LawGroupLabel and State. The data is filtered on Exclusions (Lawlevel,State) and Year. The Exclusions (Lawlevel,State) filter keeps 82 members. The Year filter has multiple members selected.

Figure 1: Q1 – Firearm Homicide Rate by State (Choropleth Map)

The second chart (Figure 2) is a **scatter plot** where each point represents one state: the X axis shows estimated gun ownership (FSS) as contextual background, the Y axis shows the total law count (`lawtotal`), and color encodes the firearm homicide rate. A **parameter** lets the viewer toggle between the quantity tier view and the law type group view (see Section 3.2), switching the color encoding across the entire chart without rebuilding it.

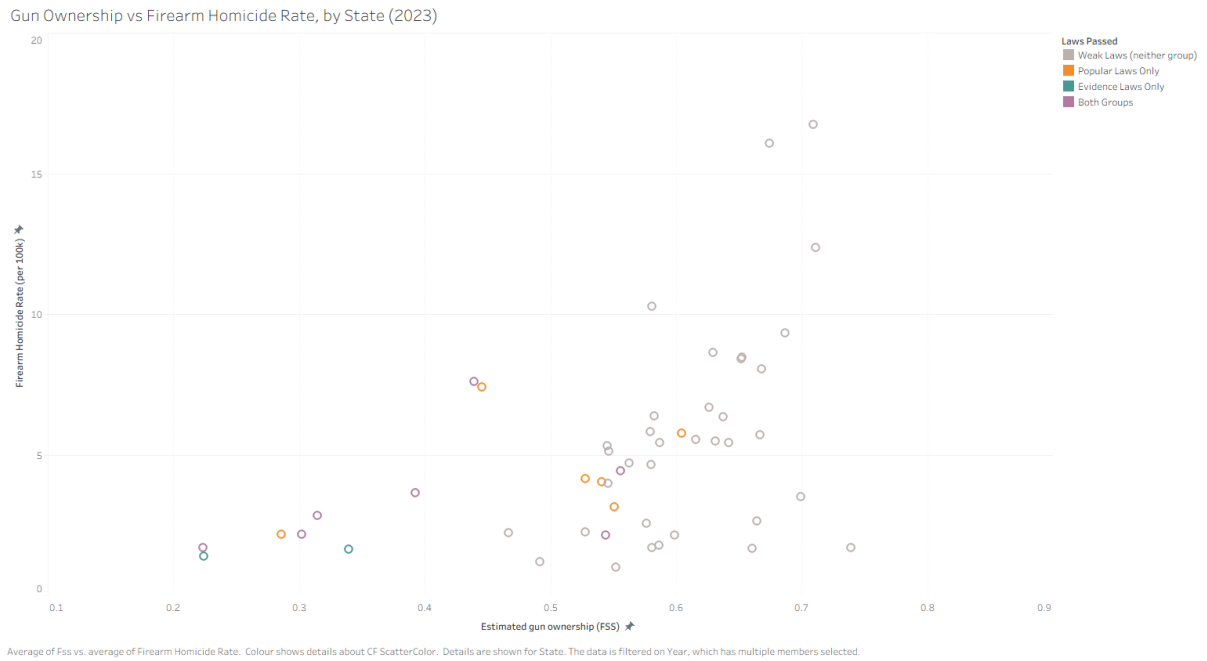


Figure 2: Q1 – Law Count vs. Estimated Gun Ownership (FSS), colored by Law Profile (Scatter Plot)

Insights: States with fewer laws cluster towards higher firearm homicide rates. The FSS axis provides ownership context but is not treated as a causal variable; the law count and law type dimensions are the primary analytical lenses.

4.2 Q2 – Homicide Trends by Law Profile Over Time

An **interactive line chart** (Figure 3) answers Q2^{2.6.2}. The X axis shows year (1976–2023), the Y axis shows firearm homicide rate, and each line represents a group of states defined by the active law-group **parameter** — shared with Figure 2 so both charts respond to the same toggle. When set to “Law Quantity Tier”, states are grouped into the Low, Medium, and High tiers; when set to “Law Type Group”, lines represent the four type categories — Weak, Popular, Evidence, and Both — as defined in Section 3.2. A vertical **reference line** marks notable legislative periods, making it possible to observe whether trajectories shifted around key policy events.

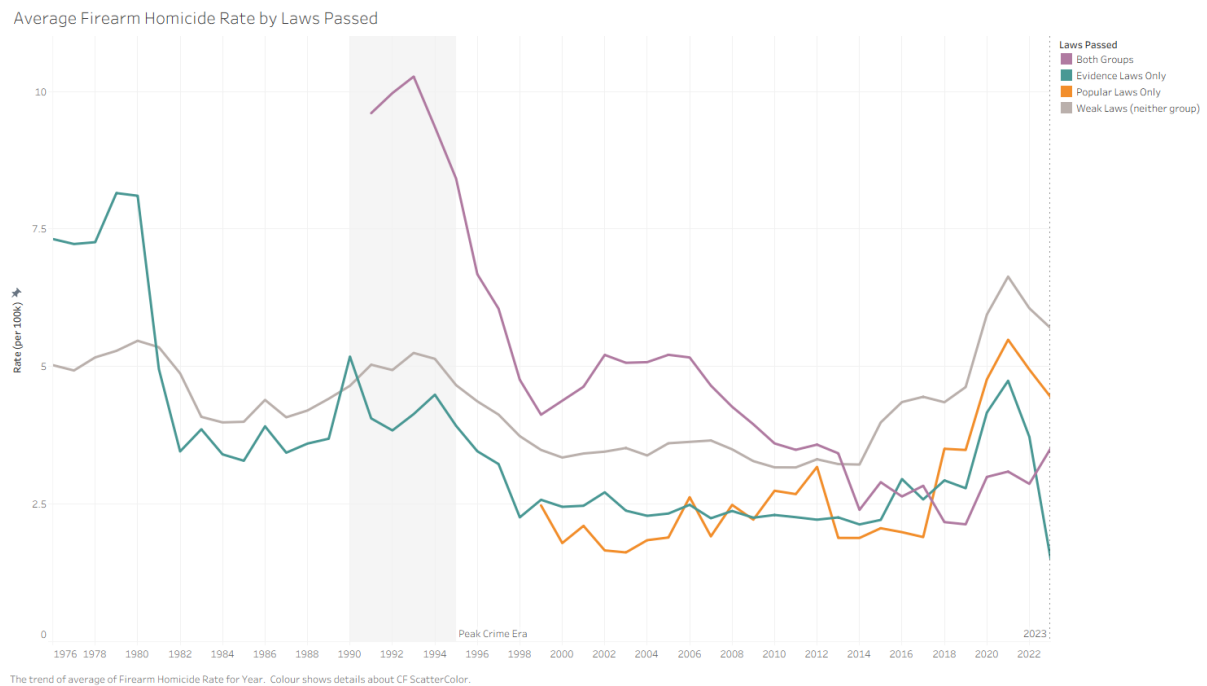


Figure 3: Q2 – Firearm Homicide Rate Over Time by Law Profile (1976–2023)

Insights: States grouped by higher law counts show a steeper decline in firearm homicide rates from the 1990s onward. The divergence between groups widens over time, and the pattern holds across both categorizations.

4.3 Q3 – State-Level Law History and Homicide Standing

Two charts answer Q3^{2.6.3}. The first (Figure 4) is a **line chart** showing the total number of firearm laws in effect over time for a user-selected state, driven by a **parameter**. This makes it easy to identify when a state underwent major legislative shifts. The second (Figure 5) is a **bar chart** ranking all states by firearm homicide rate in a selected year: bars are colored by law-group category (Section 3.2), and the selected state is highlighted using a **set** so the viewer can immediately see where it stands relative to all others. A national average **reference line** is overlaid on the bar chart to provide immediate context. Together, Figures 4 and 5 let the viewer judge whether periods of legislative growth coincide with an improvement in that state's homicide ranking.

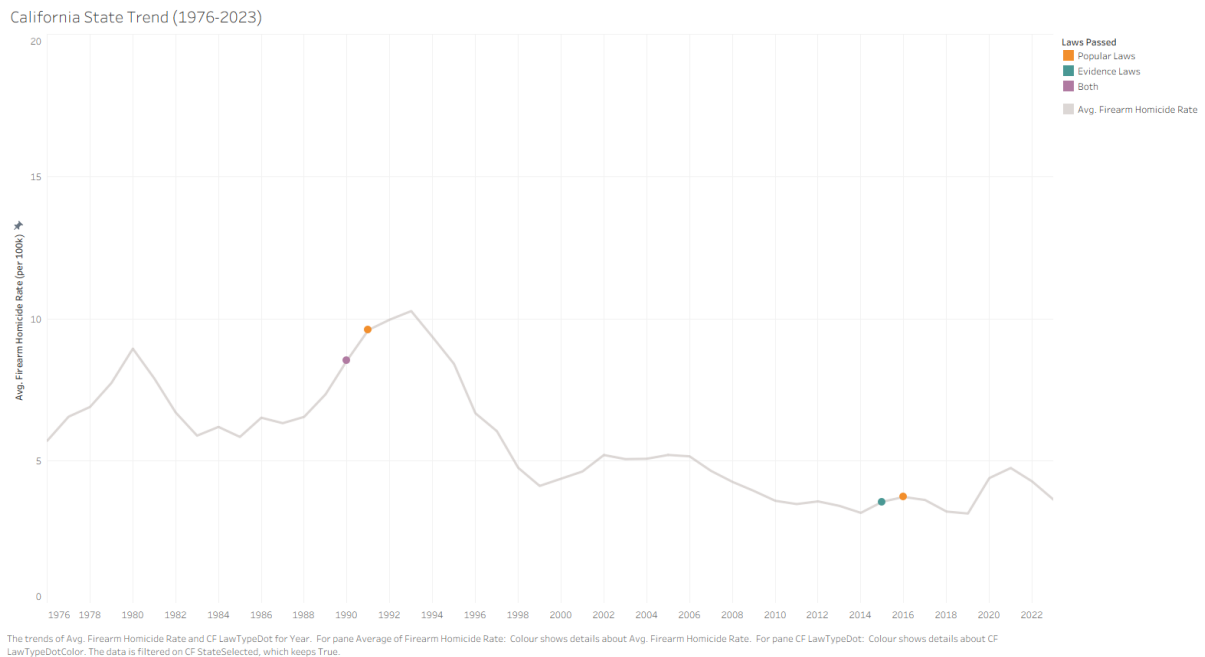


Figure 4: Q3 – Firearm Laws in Effect Over Time for Selected State (Line Chart)

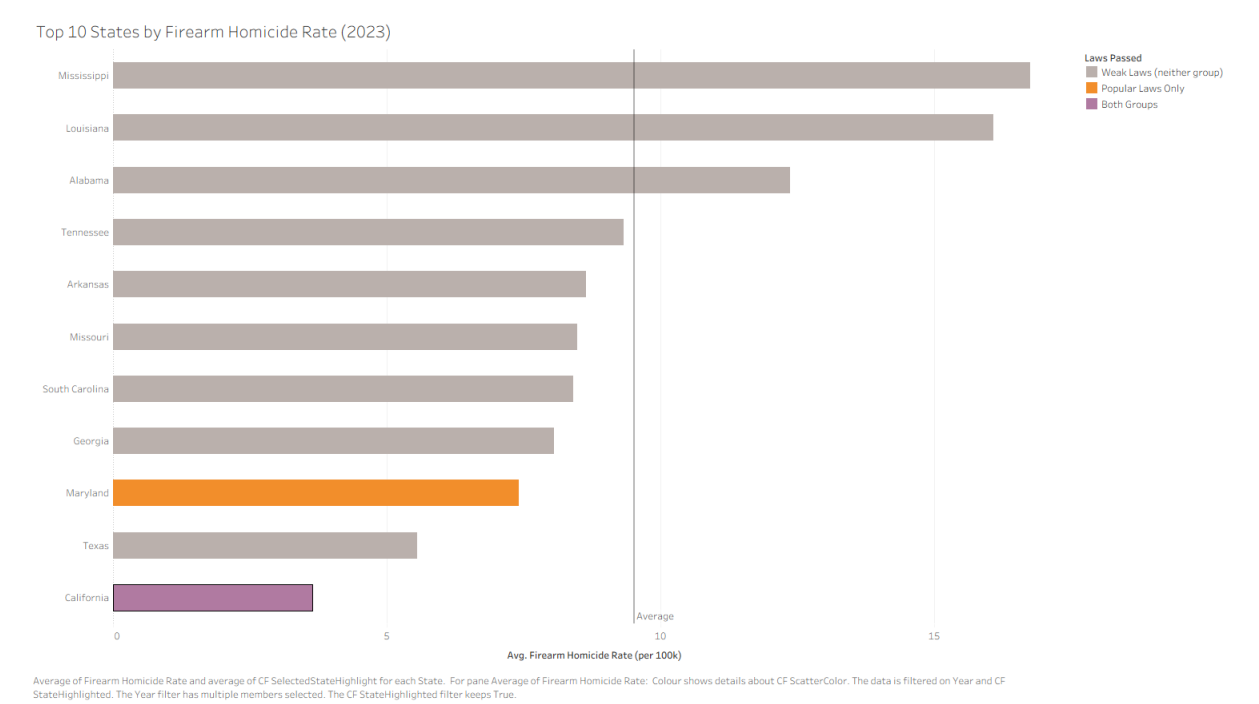


Figure 5: Q3 – Firearm Homicide Rate by State, Selected State Highlighted, National Average Reference Line (Bar Chart)

Insights: In several states, sustained increases in law count coincide with a visible improvement in homicide ranking relative to other states. The relationship is not uniform and no causal interpretation is claimed; the viewer is invited to explore individual state trajectories interactively.

4.4 Dashboards

Two dashboards integrate the individual charts into a cohesive interactive experience.

Dashboard 1 (Figure 6) combines the choropleth map (Figure 1), the scatter plot (Figure 2), and the law profile line chart (Figure 3), addressing Q1^{2.6.1} and Q2^{2.6.2} together. A **dashboard set action** on the map highlights the selected state in the scatter and updates the line chart automatically. The shared law-group **parameter** applies across all three views, so a single toggle switches the analytical lens for the entire dashboard.

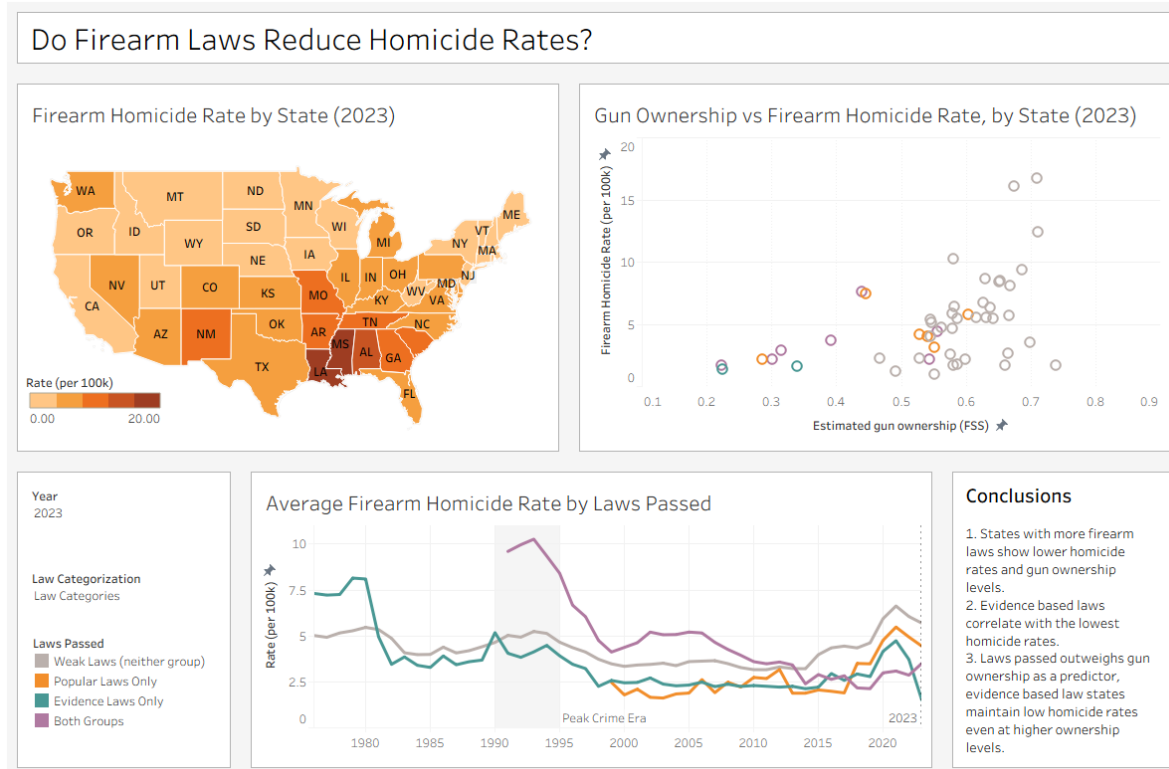


Figure 6: Dashboard 1 – Geographic Overview, Law–Homicide Scatter, and Law Profile Trends (Q1 + Q2)

Dashboard 2 (Figure 7) addresses Q3^{2.6.3} while tying it back to Q1^{2.6.1} and Q2^{2.6.2} through the choropleth map (Figure 1), the state ranking bar chart (Figure 5), and the state law history line chart (Figure 4). The shared state **parameter** drives both the bar chart highlight and the line chart, letting the viewer select any state and instantly see where it stands relative to all others and how its legislative history has evolved over time.

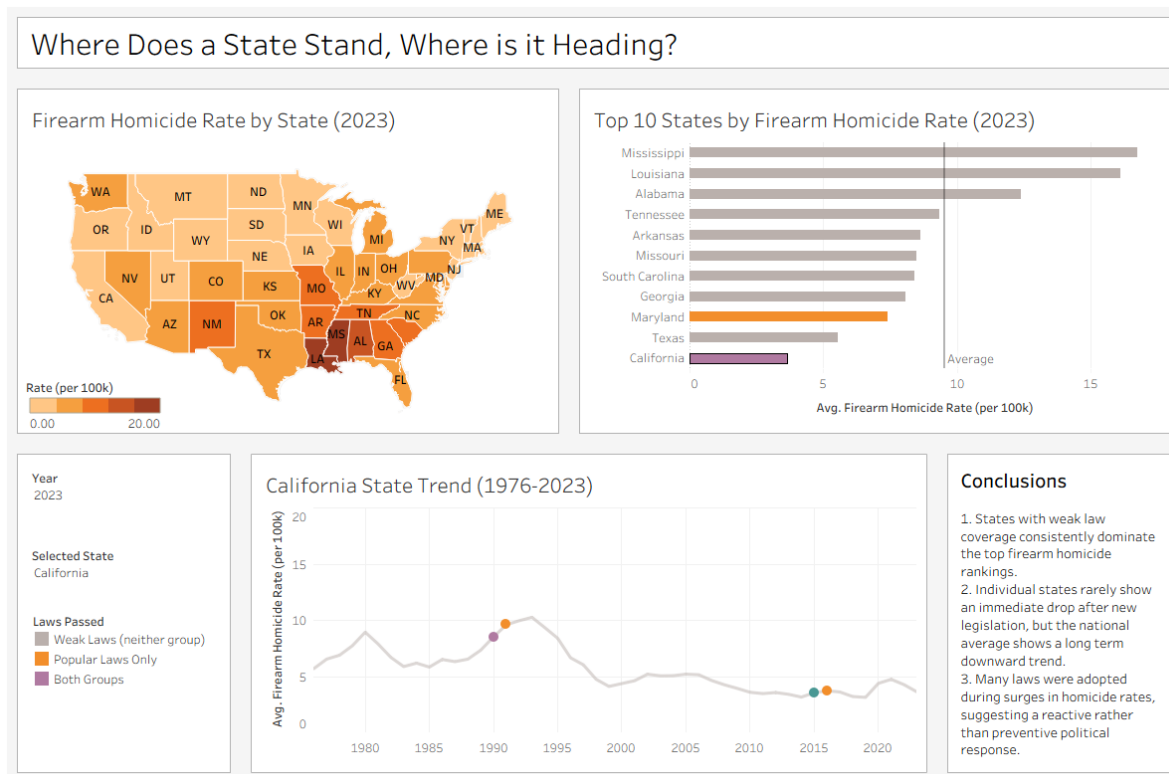


Figure 7: Dashboard 2 – Geographic Map, State Homicide Ranking, and Selected State Law History (Q3 + Q1/Q2)

4.5 Visualization Summary

Q	Chart	Chart Type	Tableau Features
Q1 ^{2.6.1}	Fig. 1	Choropleth map	Parameter, calculated field, color encoding
Q1 ^{2.6.1}	Fig. 2	Scatter plot	Parameter, color encoding
Q2 ^{2.6.2}	Fig. 3	Line chart (interactive)	Parameter, reference line
Q3 ^{2.6.3}	Fig. 4	Line chart (per state)	Parameter, filter
Q3 ^{2.6.3}	Fig. 5	Bar chart (comparative)	Set, reference line, set action
Q1+Q2	Fig. 6	Dashboard	Set action, parameter, filter
Q3+Q1	Fig. 7	Dashboard	Parameter, set action, filter

Table 2: Visualization summary

Table 2 summarizes:

- 4 chart types used across the project: choropleth map, scatter plot, line chart, and bar chart.
- Tableau features used: calculated fields, parameters, sets, dashboard set actions, reference lines (year and average), filters, and customised tooltips.
- 2 data sources joined by state and year in Tableau, in two different file formats (.csv and .xlsx).

5 Conclusions

Visualizations reveal the following:

- Q1^{2.6.1}: States with fewer firearm laws consistently show higher firearm homicide rates, both geographically and in the cross-sectional scatter view (Figure 2).
- Q2^{2.6.2}: States with more laws, whether measured by quantity or type category, have seen steeper declines in firearm homicide since the 1990s (Figure 3).
- Q3^{2.6.3}: In several states, sustained legislative growth coincides with a visible improvement in homicide ranking relative to other states, though no causal claim is made (Figures 4 and 5).
- Gun ownership (FSS) provides useful contextual framing but is not treated as a causal variable; the regulatory environment is the primary analytical lens throughout the project.

Limitations include the proxy nature of the FSS ownership estimate, the absence of statistical controls for confounding variables such as poverty and urbanization, and the use of a static join on state name which may mask within-state variation.

6 References

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